BAOMING ZHANG

Ph.D. Candidate in Computer Science | Continual Learning and Reliable Deep Networks
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RESEARCH PROFILE

Ph.D. candidate studying continual and test-time adaptation, deep-network optimization, and reliable learning under distribution shift. My research combines controlled analysis of online update dynamics with parameter-efficient adaptation for vision and multimodal foundation models.

EDUCATION

The University of Texas at Dallas, Richardson, TX **Sep 2023 - May 2028 (expected)**
Ph.D. in Computer Science (GPA: 3.8/4.0) | M.S. in Computer Science, Dec 2025 (GPA: 3.8/4.0)
Chongqing University, Chongqing, China **Jul 2021**
M.S. in Control Science and Engineering (GPA: 3.7/4.0)

RESEARCH EXPERIENCE

Graduate Researcher, UT Dallas — Sep 2023 - Present

Grounding DINO Continual Test-Time Adaptation — ***Ongoing***

- Reconstructed and audited the published Test-Time Adaptive Object Detection with Foundation Model (TTAOD-F) pipeline for Grounding DINO on 5,000 COCO-C Gaussian-noise images; prompt-and-memory adaptation increased mAP from 13.82 to 18.74 (+4.93 points) over the frozen detector.
- Designed a paired one-batch-ahead protocol that restores identical model, optimizer, memory, and RNG state before comparing adapt versus skip on the same future batch; measured 274 helpful, 267 harmful, and 708 neutral updates under a preregistered threshold.
- Using the observed update heterogeneity to study selective adaptation rules that balance plasticity and retention in open-vocabulary detection streams.

PALM — ***Continual Test-Time Adaptation, AAAI 2025 Oral***

- Co-developed a pseudo-label-free gradient criterion to select adaptable layers and a sensitivity-based mechanism that assigns layer-wise learning rates under sequential corruptions.
- Reduced mean error over LAW by 1.7/1.6/0.9 points on CIFAR-10C/CIFAR-100C/ImageNet-C while updating only 4.34%/6.65%/17.21% of model parameters per task.

Test-Time Adaptation for Foundation Models

- Co-developed BATCLIP, an online bimodal adaptation method that updates about 0.044% of CLIP parameters and improves CIFAR-10C accuracy by 6.48 points over SAR while adapting both visual and text representations.
- Built AVROBUSTBENCH to evaluate supervised, self-supervised, and online adaptation across four audio-visual datasets with 15 correlated bimodal corruptions at five severity levels.

Reliable and Data-Efficient Learning

- Co-developed LH3D, a three-stage learnability-driven active-learning method for roadside 3D detection that used 12.98% of the DAIR-V2X-I training pool and exceeded PPAL by up to 2.19 average 3D AP points.
- Co-developed SafeFix, a controlled-generation model-repair pipeline that improved test accuracy by up to 2.89 points on CelebA and raised ImageNet10 ResNet-18 accuracy from 71.73% to 74.88% without introducing new bugs.

Research Assistant, University of Missouri — Aug 2022 - Aug 2023

- Developed segmentation-conditioned 3D residual diffusion models for abdominal CT synthesis, integrating anatomical priors and evaluating image fidelity with SSIM/FID and downstream segmentation.

Research Assistant, NUS Research Institute — Jul 2021 - Jul 2022

- Developed self-supervised and generative methods for rotating-machinery fault diagnosis using wavelet transforms, multi-instance contrastive learning, ResNet, and self-attention GANs.

PEER-REVIEWED PUBLICATIONS

[1] R. Mao*, **Baoming Zhang***, N. Ruozzi, and Y. Guo. *Learnability-Driven Submodular Optimization for Active Roadside 3D Detection.* **CVPR 2026, pp. 11579-11588.**

[2] O. Xu, **Baoming Zhang**, R. Mao, and Y. Guo. *SafeFix: Targeted Model Repair via Controlled Image Generation.* **Transactions on Machine Learning Research, 2026.**

[3] D. He*, **Baoming Zhang***, X. Wang*, Y. Hao, D. Yang, and Y. Duan. *PIVM: Diffusion-Based Prior-Integrated Variation Modeling for Anatomically Precise Abdominal CT Synthesis.* **IEEE ISBI 2026 (Oral).**

[4] S. K. Maharana, S. S. Kushwaha, **Baoming Zhang**, A. Rodriguez, S. Wei, Y. Tian, and Y. Guo. *AVROBUSTBENCH: Benchmarking the Robustness of Audio-Visual Recognition Models at Test-Time.* **NeurIPS 2025 Datasets and Benchmarks Track.**

[5] S. K. Maharana, **Baoming Zhang**, L. Karlinsky, R. Feris, and Y. Guo. *BATCLIP: Bimodal Online Test-Time Adaptation for CLIP.* **ICCV 2025, pp. 1569-1579.**

[6] S. K. Maharana, **Baoming Zhang**, and Y. Guo. *PALM: Pushing Adaptive Learning Rate Mechanisms for Continual Test-Time Adaptation.* **AAAI 2025 (Oral), pp. 19378-19386.**

[7] **Baoming Zhang**, Y. Mao, X. Chen, Y. Chai, and Z. Yang. *Self-Supervised Learning Advance Fault Diagnosis of Rotating Machinery.* **NCAA 2021, pp. 319-332.**

[8] X. Chen, Y. Mao, **Baoming Zhang**, Y. Chai, and Z. Yang. *A Method for Imbalanced Fault Diagnosis Based on Self-Attention Generative Adversarial Network.* **NCAA 2021, pp. 333-346.**

** Equal contribution*

TEACHING EXPERIENCE

Teaching Assistant, UT Dallas **2023 - 2026**

Courses: Database Systems (Fall 2023, Spring/Fall 2025); Operating Systems Concepts (Spring 2024); Programming Languages (Fall 2024, Spring 2026); Machine Learning (Summer 2026); Artificial Intelligence (Fall 2026).

TECHNICAL SKILLS

Research: Continual learning, test-time adaptation, optimization, distribution shift, model reliability, active learning
Programming: Python, C++, MATLAB, Java, SQL
Frameworks and tools: PyTorch, PyTorch Lightning, Hugging Face, OpenCV, TensorFlow, scikit-learn, Docker, Git, LaTeX
Models and methods: Grounding DINO, DINOv2, CLIP, diffusion models, online adaptation, controlled experiments, bootstrap analysis
Open-source projects: LH3D (CVPR 2026) | PIVM (ISBI 2026 Oral)

ACADEMIC SERVICE

Reviewer: ECCV 2024/2026; CVPR 2025/2026; NeurIPS 2026; ISBI 2026; ICIP 2026; BMVC 2026; Medical Physics